

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Static Var System Templates

WECC SVSMO1 100/-100/20/-140Mvar 50Hz,
WECC SVSMO1 100/-100/20/-140Mvar 60Hz,
WECC SVSMO2 100/-100/20/-140Mvar 50Hz,
WECC SVSMO2 100/-100/20/-140Mvar 60Hz,
WECC SVSMO3 10Mvar 50Hz,
WECC SVSMO3 10Mvar 60Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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1 Introduction

This document describes the usage of the WECC SVSMO1, SVSMO2 and SVSMO3 Static VAr System (SVS) models as implemented in *PowerFactory*. The models are available as "General Templates" in the *PowerFactory* global library for enabling easy model deployment in any existing *PowerFactory* project. The model is based on the "WECC - Generic Static Var System Models for the Western Electricity Coordinating Council" guideline [1]. This document describes the usage of these models in *PowerFactory* and it highlights details which either deviate or are not specified in [1]. For a better understanding of this document, the reader is encouraged to first refer to the WECC guideline as this document does not intend to provide in-depth details on the implementation of the models but rather the specifics of the *PowerFactory* model implementation.

The following SVS models according to WECC [1] are available in the *PowerFactory* global library:

- SVSMO1
- SVSMO2
- SVSMO3

Chapter 2 provides information on the quick use of the developed models. general description of the SVS models developed based on WECC guidelines. Chapter 3 provides a high level description of the most relevant functionality of the available simulation model with focus on the *PowerFactory* specific implementation. Chapter 4 guides the user on the most common configuration tasks that may be needed in order to customise the SVS models such that they correspond with a specific SVS installation setup.

Note: The models which are available in *PowerFactory* as templates are configured with a set of default parameters. The user should consider adjusting these parameters (i.e. power rating, voltage level, parameters of the dynamic models, etc) to represent the actual equipment specifications being subject to the respective analysis.

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2 SVSMO1/SVSMO2/SVSMO3 Model - Quick Use Instructions

The SVSMO1, SVSMO2 and SVSMO3 models consist each of a step-up transformer, a number of shunt elements and the corresponding low- and high-voltage buses. The SVSMO1 and SVSMO2 are represented by one SVS element (Static Var System *ElmSvs*), while for the SVSMO3 the static generator (*ElmGenstat*) was employed. The models are delivered as templates for the purpose of simple network deployment and ease of use.

A new model (based on a template) can be added to an active project as follows:

- From the single line diagram, make sure that the “Freeze” mode is disabled, i.e. one can drop in the diagram new network elements.
- Press the *General Templates* button available in the drawing toolbar;
- A list of available templates is shown. Select the one you want to use;
- Left click (n.b. single left mouse click!) anywhere into the single line diagram. The network elements corresponding to the template appear and are bound to the mouse icon. The model is not yet deployed in the network;
- To insert it into the single line diagram, position the mouse icon (with the bounded elements) in the diagram area where the model should be deployed and left click once more;
- The process of deploying new models could continue by clicking again (n.b. left click) in the single line diagram.
- Once all models are created, the template drawing mode can be exited by either closing the template window or by right click in an empty area of the single line diagram.

Note: It is **NOT** recommended to use a “Copy&Paste” approach to replicate the model since this would create only a copy of the grid element(s) (and not of the dynamic controller(s) or any other element(s) not visible (hence selected) from the single line graphic).

Should you have already operated changes on a specific unit and wish to re-use the same settings for other identical equipment then you may create a new template based on which new models can use it as source data. Hence, please use the template function to copy the model or to bring it into another project. For this purpose, follow the steps below:

- Mark all elements belonging to the equipment you want to duplicate. This may also include a transformer. Remind to mark all elements where measurement devices are connected to. The SVS models are using the HV bus of the transformer as measurement location.
- Right click now at the marked elements and select “Define template”.
- Select a meaningful name and press “OK”.
- Confirm in the next step with “Yes” that you want to include external objects (the external object is the composite model with all controllers and measurement devices).
- Make also sure that any load flow control element (e.g. Shunt Controller elements) are also included in the template (if not already done so)
- You can use now the just created template in the active project as described above.

There are several additional steps necessary to move the above created template to another project:

- The above create template is stored in the project library in the sub-folder “Templates”. Open this folder in the data manger.
- Right click then at the previously created template and select “Edit”.
- Press in the following window the “Pack” button. This will copy all types of the model to the template.
- The template can now be copied to any other project (to the Template folder of the project library) or exported as “*.pfd” file.

3 SVSMO1/SVSMO2/SVSMO3 Models

3.1 SVSMO1 Model Overview

The model single line diagram is shown in Figure 3.1.

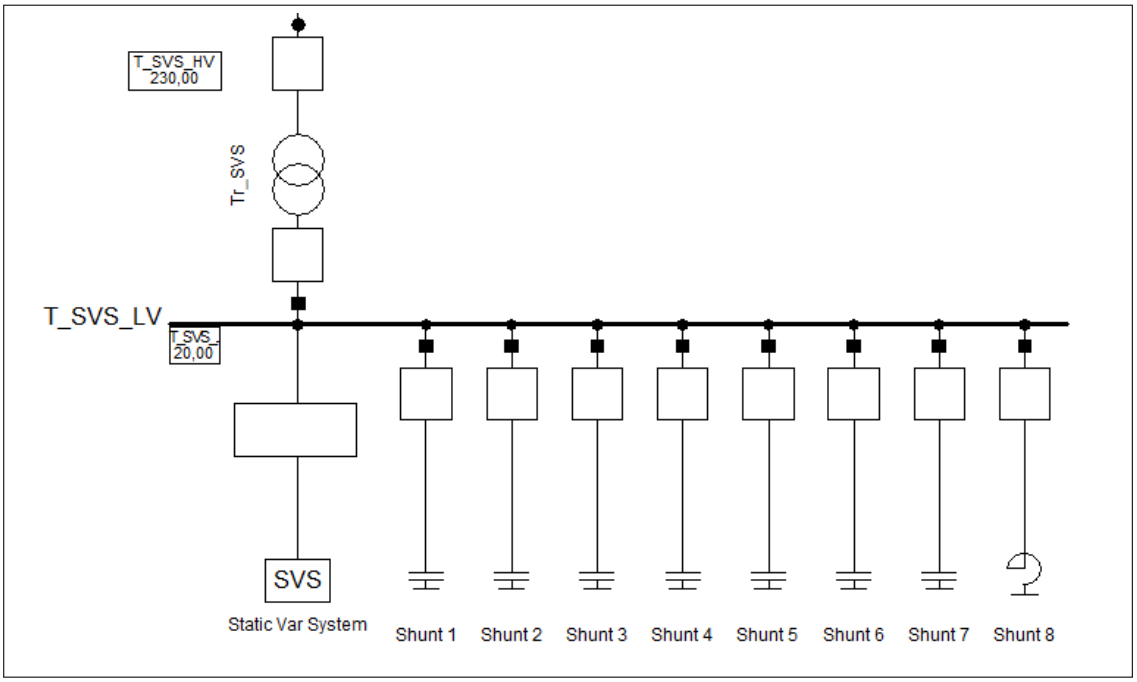


Figure 3.1: SVSMO1 Single Line Diagram

The model control block diagram is shown in Figure 3.2.

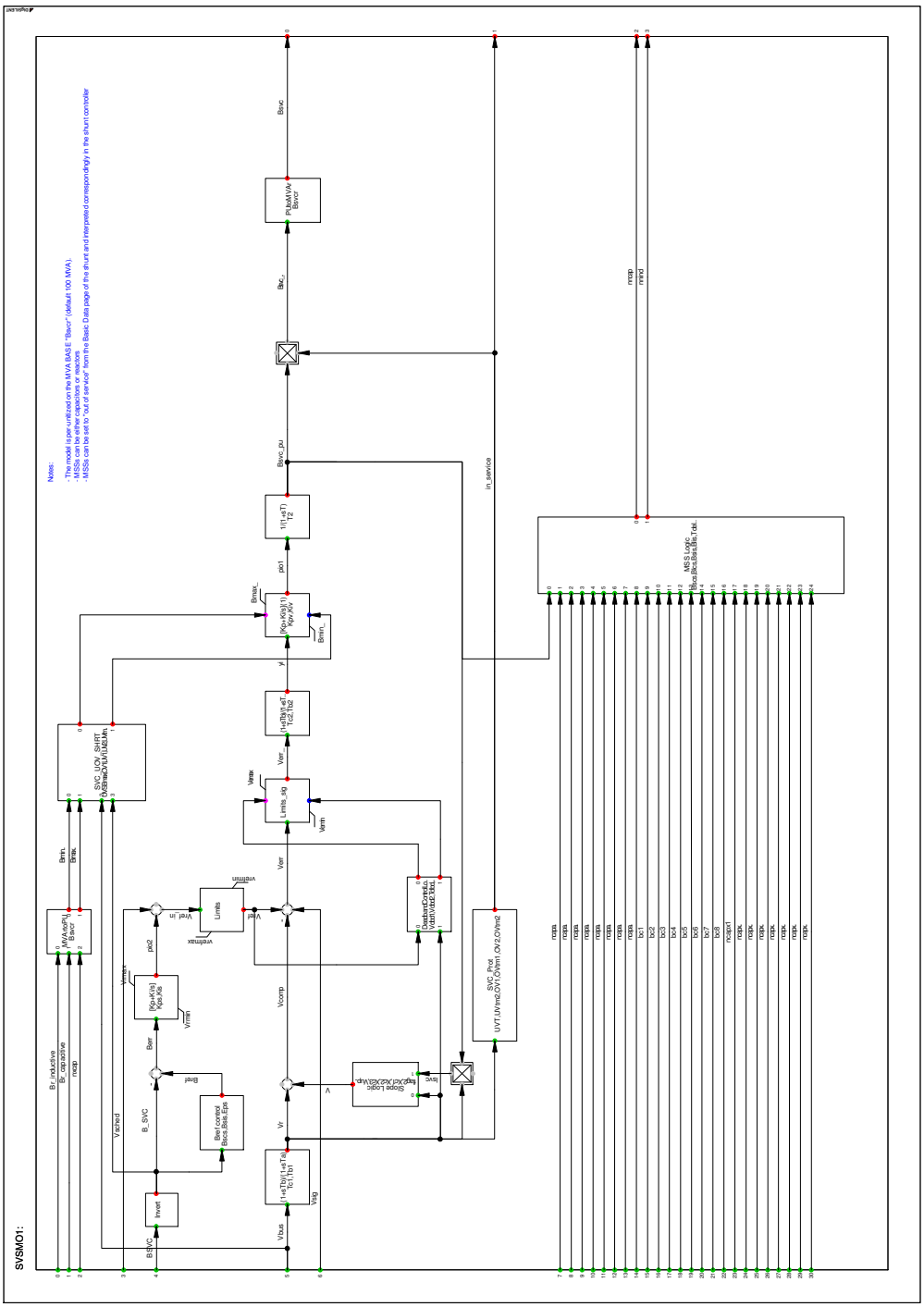


Figure 3.2: SVSMO1 Control Diagram

The SVSMO1 and SVSMO2 models are delivered as templates and are located within the *PowerFactory* global library (Templates\FACTS). The ratings of the available templates (named

WECC_SVSMO1_100/-100/20/-140Mvar_50Hz, *WECC_SVSMO1_100/-100/20/-140Mvar_60Hz*, *WECC_SVSMO2_100/-100/20/-140Mvar_50Hz* and *WECC_SVSMO2_100/-100/20/-140Mvar_60Hz*) are given below:

- Nominal voltage, LV side: 20 kV
- Nominal voltage, HV side: 230 kV (this voltage can be easily adapted by adjusting the step-up transformer ratings and the corresponding HV bus)
- TCR (thyristor controlled reactor): 100 MVAr (build-in the Static Var System *ElmSvs* element)
- TSC (thyristor switched capacitor): 100 MVAr (build-in the Static Var System *ElmSvs* element)
- MSC (mechanically switched capacitor): 140 MVAR within 7 units. Each unit consists of one shunt element with two steps and 10 MVAr rating per step. The MSCs are explicitly defined using shunt elements (not build-in the “ElmSvs” element)
- MSR (mechanically switched reactor): 20 MVAR within 1 unit. There is one reactor represented via one shunt element with two steps and 10 MVAr rating per step. The MSR is explicitly defined using shunt elements (not build-in the “ElmSvs” element)

Note: The above configuration can be customised according to the user requirements by following the instructions in this User Manual.

The relevant functions of the SVS models (for both SVSMO1 and SVSMO2) are:

- Voltage regulator
- Coordinated switching logic for MSSs
- Slow susceptance regulator
- Deadband control
- SVC slope/droop functions
- SVC limits
- SVC Over- and Under-Voltage Strategy including tripping function
- Short term rating capability

The model is designed for use in analysis types explicitly intended by the WECC Task Force as described in [1]:

- Load Flow calculation and
- Stability (RMS) simulation

Hence, after model configuration (refer to Section 4), the two above mentioned studies can be executed without any further model modifications/customisations. For using the model with any other calculation functions, appropriate considerations must be undertaken in order to obtain realistic behaviour. The model equations and control diagrams are open and available for inspection, hence the user can apply further modifications to it, if so required.

3.2 SVSMO2 Model Overview

Model usage and configuration is almost identical with the information provided for the SVSMO1 model (see previous section). Specific to the SVSMO2 is the customisation of the deadband parameter “dbb” and of the discrete susceptance characteristic “array_Bsvs”, available in the edit dialog of the common model “SVSMO2”. The default characteristic is shown in Figure 3.3. The deadband and the characteristic may be customised as needed.

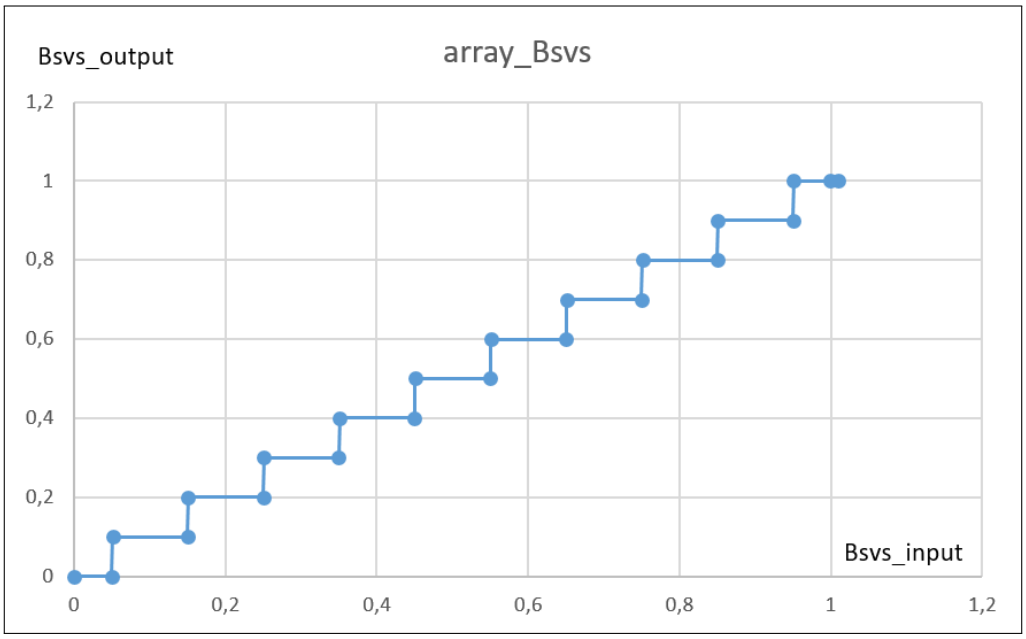


Figure 3.3: SVSMO2 lookup array for discrete susceptance control (default configuration)

3.3 SVSMO3 Model Overview

SVSMO3 model is very similar to SVSMO1 being the main difference between the two models that SVSMO3 implements a voltage source converter (VSC) based SVS, in this case a STATCOM. The model single line diagram is shown in Figure 3.4.

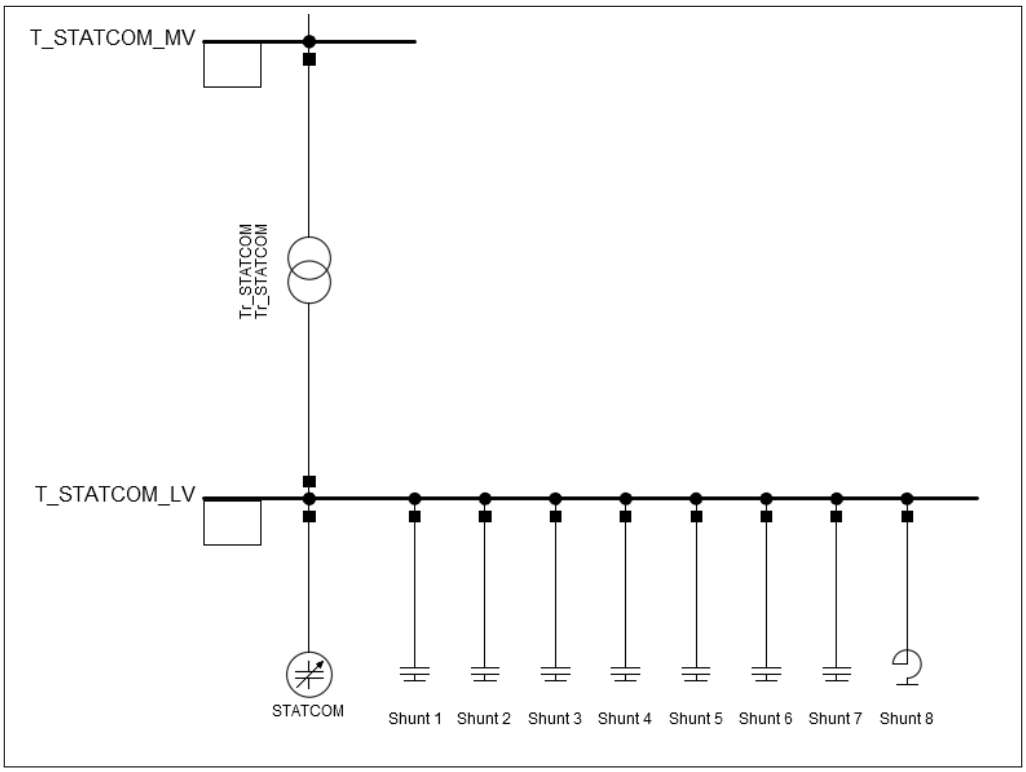


Figure 3.4: SVSMO3 Single Line Diagram

3.4 Model Usage Guidelines

3.4.1 Load Flow Analysis

In load flow calculation, the SVS element (*Static Var System.ElmSvs*) has the purpose of regulating the reference bus voltage (by default this is the HV side of the step-up transformer). The model includes a shunt control element (*Shunt Controller.ElmShntcntrl*) that controls the tap position of the mechanically switched shunts (MSS) (if any). The purpose of this controller is to minimize the reactive power output of the SVS element, such that in steady state the SVS element will always maintain sufficient control band in both inductive/capacitive operation modes. The control band of the shunt controller is adjusted to match the steady state capability of the unit, which is equivalent to the parameters “Bscs” and “Bsis” of the dynamic model (*ElmDsl*), as shown in Figure 3.5.

Note: The full capability, i.e. dynamic rating, of the SVS (TCR and TSC) is defined by the dynamic model parameters “Blcs” and “Blis” (which, in the delivered template are +/- 100Mvar). The steady state capability is defined by the parameters “Bscs” and “Bsis” (which, in the delivered template are +/- 20Mvar). The load flow shunt controller (*ElmShntcntrl*) will emit a warning in the output window as soon as the SVS unit (*ElmSvs*) will exceed the steady state bounds even if the full capability has not yet been reached. Whenever such a warning is generated, the SVS operates normally but the MSS unit reached either the maximum or minimum tap positions.

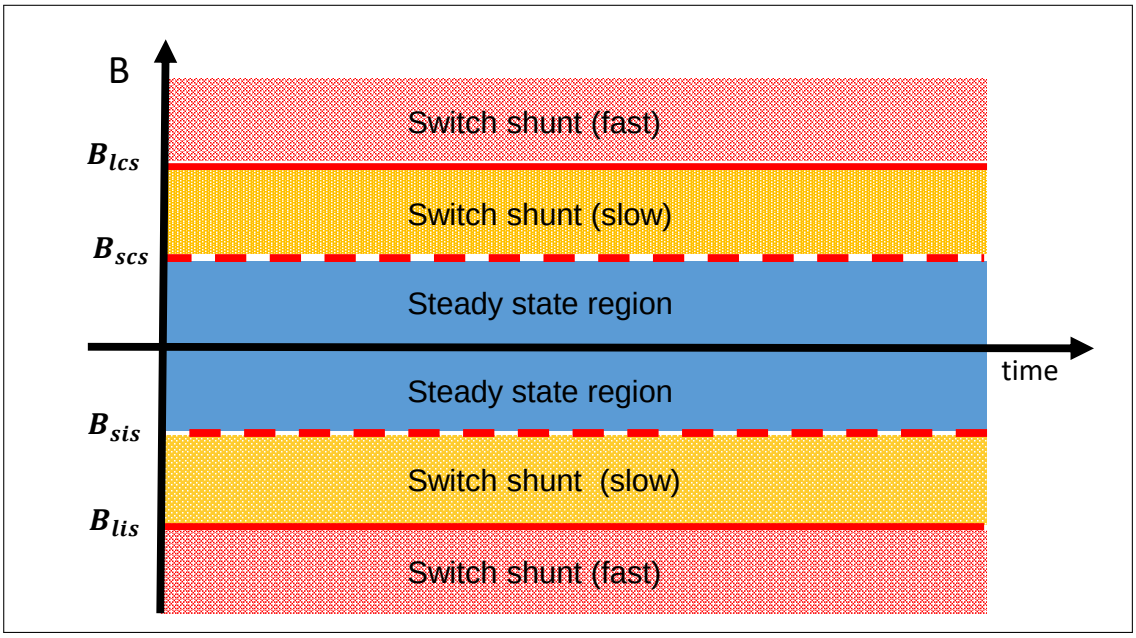


Figure 3.5: SVSMO1 shunt control region

For the SVSMO3 model, a station controller element (*ElmStactrl*) commands the STATCOM to regulate the voltage at the reference bus voltage (like in the other two models, this is the HV side of the step-up transformer). At the same time a shunt controller intends to regulate the output of the STATCOM such that this has enough remaining control margin.

3.4.2 Dynamic RMS(Stability) Simulation

The model is appropriate (and has been tested) for RMS dynamic studies (balanced/unbalanced) considering the following simulation specifications:

- Maximum time step: 0.01 s (10 ms)
- Minimum time step: 0.001 s (1 ms)

Note: The above specification recommends that the simulation time step settings should be within the forementioned range. There is no known restriction in terms of the fixed/variable time step option, i.e. both fixed and variable simulation step size adaptation settings are possible.

Other notes:

- The model operates on positive sequence quantities (feedback signals as well as set-points). The model can be used within an unbalanced RMS simulation, nevertheless, the default operation will remain, surely, as a positive sequence control scheme.
- In case non-zero lead time constants are introduced, i.e. parameters “Tc1” or “Tc2”, then in order to improve numerical stability at relatively large simulation time steps, it may be

needed, depending on the particular case, that the models are set to operate with a “A-stable integration algorithm”. For this purpose, proceed as below:

- Open the SVS element (*Static Var System.ElmSvs*), go to the RMS-Simulation page and set the flag “A-stable integration algorithm”;
- Open the “SVSMO1” (or “SVSMO2”) common model and set the flag “A-stable integration algorithm”;
- Open the voltage measurement device “Vmea_SVS” and set the flag “A-stable integration algorithm”.

4 Model Configuration

This chapter contains information about the configuration of the different SVS models. In section 4.1 and 4.2, the general configuration of SVSMO1/SVSMO2 models and SVSMO3 respectively is described, for example, the adjusting of the rated power or changing the mechanically switched capacitor settings.

4.1 SVSMO1 and SVSMO2

4.1.1 Adjusting Rated Power

The rated power is defined via the rating parameters located in:

- the SVS element (*Static Var System.ElmSvs*): adjust here the TCR and the TSC ratings; leave the MSC ratings to zero, as the model is designed to consider external mechanically switched shunts
- the shunt elements: they can be either capacitors or reactors; adjust for each shunt the type (e.g. C or R-L), the nominal voltage, rated reactive power, per step, and the maximum number of steps.
- the SVSMO1/SVSMO2 dynamic model:
  - open the edit dialog of the common model SVSMO1 (or SVSMO2).
  - adjust the parameters “Bscs”, “Bsis”, “Blcs” and “Blis” accordingly. Refer to WECC documentation for details [1].

4.1.2 Adjusting load flow control setpoints

The load flow control is configured in the SVS element (*Static Var System.ElmSvs*) as such:

- Open the load flow page of the SVS element edit dialog
- Observe the “Remote control” flag and the “Controlled Node” reference busbar. This busbar is controlled by the SVS unit.
- The “Voltage Setpoint” is programmed in the corresponding field.
- Observe the programmed reactive power droop characteristic in the “Enable Droop” pane. The settings are similar with the droop control settings from the dynamic controller

4.1.3 Adding or removing shunt elements from the SVS system

With the SVSMO1 (or SVSMO2) models, it is possible to use any number of shunt elements between zero and eight (included). The shunt elements can be either capacitors or reactors. Each shunt can be set either with a single step or with multiple steps. When adding or removing a certain shunt from the setup, the user should keep in mind the following considerations:

- Update (add or remove) the shunt element of your choice
- Update accordingly the Shunt Controller element (ElmShntcntrl) that operates the MSSs in load flow calculations: open its edit dialog and update the shunt list in the “Load Flow” page. Also, make sure the control settings are still valid for your particular installation.
- Update the dynamic model, i.e. the composite model “Model_SVS”, as below:
  - when **removing** a shunt from the setup, the composite model will still link to the removed/deleted element. Make sure to reset the connection, such that the corresponding field is blank (see Figure4.1).
  - when **adding** a shunt to the setup, the composite model will not link automatically the newly added element. This process must be done manually. Make sure that the new element is linked in the reset the connection, such that the corresponding field is blank.

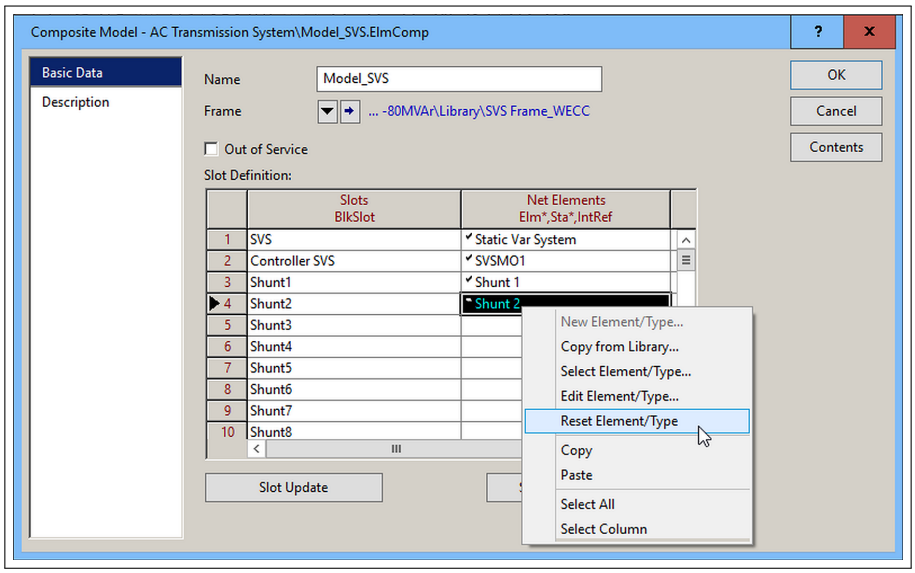


Figure 4.1: Slot update of the dynamic composite model

4.2 SVSMO3

4.2.1 Adjusting Rated Power

The rated power is defined as for the other models via the rating parameters located in:

- the STATCOM element (Static Generator *ElmGenstat*): modify here the nominal apparent power and adjust accordingly the despatch and operational limits settings
- the shunt elements: are adjusted in the same way as explained for SVSMO1/SVSMO3

4.2.2 Adjusting load flow control setpoints

The load flow control is configured via the following elements:

- Open the load flow page of the Station Control element (*Stactrl*)
- Notice the ”Controlled Node” reference bus bar and the ”Voltage Setpoint” setting, this is controlled by the STATCOM

4.2.3 Adding or removing shunt elements from the SVS system

The configuration for the SVSMO3 is identical to models SVSMO1 and SVSMO2 described before.

4.3 Dynamic Model Configuration

The dynamic simulation is based in *PowerFactory* on the load flow calculation. This means that the starting point for the dynamic simulation will always be the load flow solution. It is therefore important to check that the load flow configuration of the elements is correct.

The composite model which contains the links to all dynamic model parts (common models) can be accessed via the Data/Model Manager.

- Via Data or Model Manager: Search the corresponding composite model , right click and select “Edit”.
- After opening the composite model the different components of the SVS model can be accessed via a double click in the second column.

Note: The models which are available in *PowerFactory* as templates are configured with a set of default parameters. The user is encouraged to adjust all relevant parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic model) to represent a specific unit.

The dynamic (common) models can be configured via scalar input parameters. These parameters are displayed after opening a common model on the “General” page. Some models (e.g. SVSMO2) may contain array parameters (i.e. a list of parameters). The arrays can be found on the second tab of the model edit dialog (i.e. the common model), tab named “Advanced 1”. This is the case of the SVSMO2 model which implements a discrete susceptance characteristic based on an array parameter.

To change the dimension of an array double click at the cell in the first column. You will be allowed to introduce there the number of rows within the array. Switch back and forth between the “General” and the “Advanced 1” tabs and the table will be automatically updated with the requested number of rows. In the example shown in Figure 4.2, an array of 24 rows has been predefined, which represents a stepwise characteristic of the discrete susceptance output.

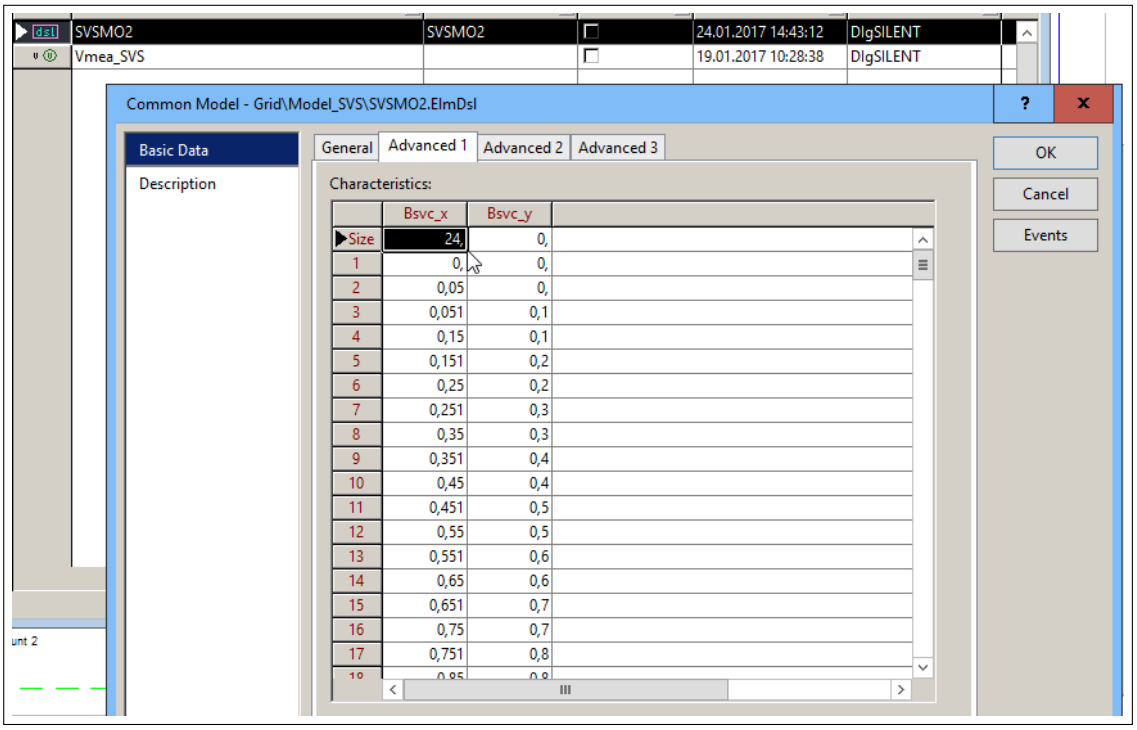


Figure 4.2: Array input of parameter array_Bsvs

5 Model Documentation

The model is thoroughly described in the WECC reference [1]. Please refer to the guideline documentation for a detailed description of the dynamic model.

6 References

[1] WECC - Generic Static Var System Models for the Western Electricity Coordinating Council; Issued april 18, 2011, 2011.

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